Durrow Creek Bridge 2209-04, Ransom Bar Road: the 185 ft span carries the design flood

Hydrology report for the WSDOT Local Programs structures review. Prepared by Amy Rasmussen, PE, Bridge Engineer. 6 November 2026. Subject: design flood and span selection, Ransom Bar Road over Durrow Creek (County Road 2209, milepost 4.31).

1. Recommendation

Build the 185 ft span. Durrow Creek's 1 per cent annual exceedance flood at the bridge is about 25,000 cfs, which the 185 ft span carries within its rated 29,000 cfs; the 140 ft span, rated 20,500 cfs, does not. The 240 ft span is not needed.

2. Basin and records

Durrow Creek at the Ransom Bar Road crossing drains 85.3 square miles of foothills between 610 and 4,380 ft (mean 1,870 ft). The creek has never been gauged. Under section 7.4.3(b) of the County standard the record carried to the site is that of the nearest gauged basin with 25 years or more of annual peaks: the Coyner River near Tessel (USGS 12183450), 86.4 square miles, 6.2 miles away, with 75 annual peaks from water year 1951 to 2025 and its peaks above the 2,500 cfs base.

3. Carrying the Coyner River's record to the creek

The two basins are the same size but not the same height. The Coyner River drains the flank of the volcano up to 10,380 ft (mean 6,350 ft); in a winter storm only its area below the snow line drains rain, and above that line the storm's precipitation goes into the snowpack. Durrow Creek's basin lies below the snow line in nearly every storm. Each Coyner River flood was therefore carried to the creek at the ratio of the storm rain each basin took below that storm's snow line, not at the ratio of the drainage areas.

For each of the 183 floods in the river's annual and above-base peak files fed by rain, the storm is the run of wet hours in the Tessel Field Airport record (the routine hourly reports) leading to the peak. Its snow line was placed from the airport's temperature while the rain fell, weighted by the hour's rain, with the lapse rate the Tessel and Durrow Ridge stations show on days both catch 0.5 in or more, 5.4 degrees C per kilometre (2,158 days; the Hatchet Meadow and Corbel Ridge pair gives 5.2), and the temperature at which half the precipitation falls as rain at the two SNOTEL pillows, 0.9 degrees C. Precipitation rises about 10 per cent per 1,000 ft (Durrow Ridge's totals run 1.16 times Tessel's in the storms warm enough for both to catch rain). In those floods the river drained rain over a median 50 per cent of its basin (61 per cent in its largest tenth) and the creek over 100 per cent of its own. The river's own peaks follow the rain over its area below the snow line (correlation 0.76 in logarithms, 183 floods) much more closely than the storm's rain over the whole basin (0.55).

The construction was checked on Selby Creek near Hollis (USGS 12186050, 62 square miles of foothills like the creek's, a crest-stage record of 14 years, too short to carry under section 7.4.3): carried from the river's concurrent floods in the same way, its 11 recorded floods are reproduced to a median +5 per cent, where the drainage-area transfer runs -26 per cent and a snow line placed with the standard atmosphere's lapse rate +47 per cent.

The creek's flood in each water year is the largest of its carried floods. Water year 1997 holds no flood fed by rain in the river's files, only a spring melt peak, and enters the fit as below the low-flood threshold (4,600 cfs). Bulletin 17C was fitted by expected moments; the Multiple Grubbs-Beck test flags 2 low floods, and the station skew (-0.08) weighted with the Region 3 regional skew (-0.12) gives -0.09.

4. Flows at the bridge

Annual exceedance probability	Durrow Creek at Ransom Bar Rd (cfs)	Coyner River near Tessel (cfs)
50 per cent (2-year)	10,000	5,800
1 per cent (100-year), the design flood	25,000	18,900

Table 1. The creek's flows to the nearest 5,000 cfs, the precision the carried record supports; the Coyner River's Bulletin 17C flows to the nearest 100 cfs.

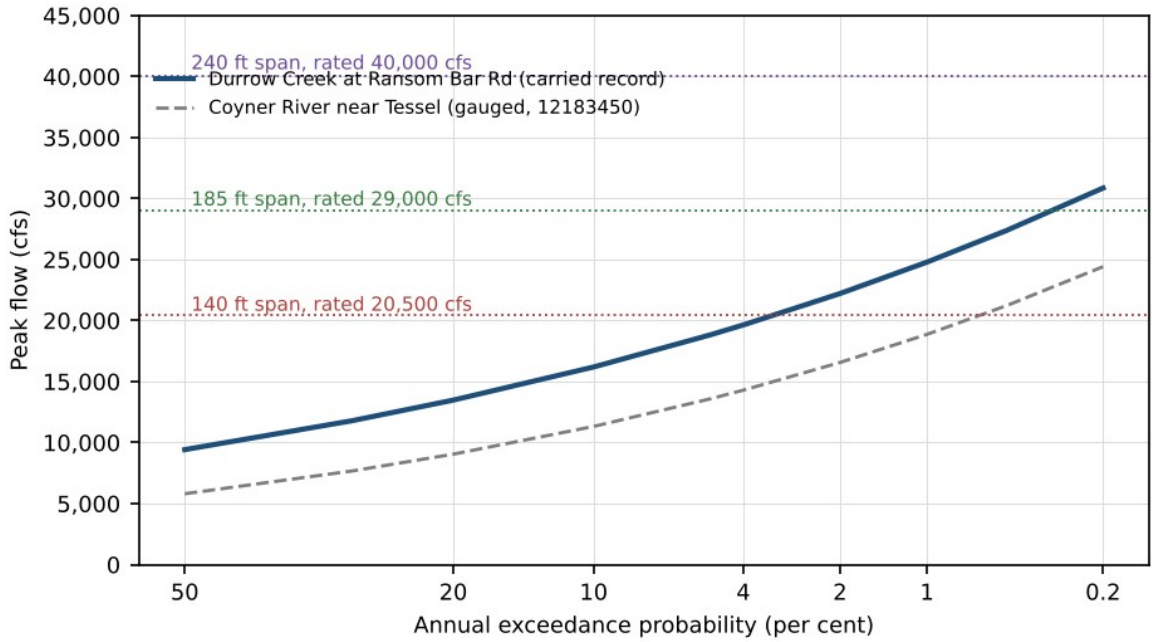


Figure 1. Frequency curves and the rated flows of the span alternatives (Hess Hydraulic Engineering, TM 26-118).

5. Span selection under section 7.4.2

Span (ft)	Rated flow, 2.0 ft freeboard (cfs)	How often the rated flow would be exceeded	Carries the design flood
140	20,500	once in 10 to 100 years	no
185	29,000	once in 100 to 1,000 years	yes
240	40,000	less often than once in 1,000 years	yes

6. Flood history at the crossing

The creek's two largest floods since 1951 in the carried record came in water years 2008 and 2011, each well above the next largest. They are not the Coyner River's largest: the river's record peaks (water years 1983 and 2001) came in storms warm enough to rain over most of its basin, while these two storms left much of the river's basin under snow and rained over all of the creek's. The estimated annual peaks are filed with this report as durrow_creek_estimated_annual_peaks.csv.

Water year	Storm (Coyner River peak)	Coyner River peak (cfs)
2008	29 January 2008	10,500
2011	27 March 2011	13,800

7. Checks

Carried by drainage area alone, the Coyner River's 1 per cent flood (18,900 cfs) gives the creek about 18,700 cfs, and that transfer reproduces the region's other gauged basins closely, because every one of them is a high basin like the Coyner River's; on it the 140 ft span would have been selected. The state's Region 3 equation is not used (section 7.4.3(c)). A snow line placed with the standard atmosphere's 6.5 degrees C per kilometre sits too low in storms and gives about 35,000 cfs; leaving out the rise of precipitation with elevation gives about 30,000 cfs; the valley's daily mean temperature, or the airport's reading at the river's peak hour (no rain fell in that hour in 48 per cent of the floods), reads air the rain did not fall in. The design flood holds at 25,000 cfs for every storm definition and rain weighting the hourly record allows.